



Semester One Examination, 2024

Question/Answer booklet

MATHEMATICS APPLICATIONS UNIT 1

Section Two: Calculator-assumed

SOLUTIONS

WA student number: In figures

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In words

Circle your Teacher's Name: Mr Bartoshefski Mrs Fraser-Jones Ms Horley Dr Kang
Ms Lastari Ms Nathan Mr Singh

Time allowed for this section

Reading time before commencing work: ten minutes
Working time: one hundred minutes

Number of additional
answer booklets used
(if applicable):

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Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet
Formula sheet (retained from Section One)

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up to three calculators, which can include scientific, graphic and Computer Algebra System (CAS) calculators, are permitted in this ATAR course examination

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	52	35
Section Two: Calculator-assumed	10	10	100	98	65
Total					100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
3. You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
4. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
5. It is recommended that you do not use pencil, except in diagrams.
6. Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section Two: Calculator-assumed**65% (98 Marks)**

This section has **ten** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 100 minutes.

Question 8

(8 marks)

Jian wants to transfer 1100 Australian dollars to his sister who lives in Singapore and will carry out the transaction using his own bank (ANZ), PayPal or Wise. The table below shows the exchange rate (how many Singapore dollars that one Australian dollar will buy) and the transfer fee charged by each company in Australian dollars. Jian will pay the transfer fee upfront to ensure that his sister receives the full amount of the transfer in Singapore dollars.

	Own bank (ANZ)	PayPal	Wise
Exchange rate	0.8464	0.8366	0.8750
Transfer fee	\$9.00	5% of transfer*	\$0.83 + 0.55% of transfer

* A minimum transfer fee of \$0.99 and maximum transfer fee of \$5.99 applies for PayPal.

- (a) Which company will charge Jian the least transfer fee? Justify your answer. (3 marks)

Solution
ANZ fee: \$9.00.
PayPal fee: $1100 \times 0.05 = 55 \rightarrow \5.99 (capped).
Wise fee: $0.83 + 1100 \times 0.0055 = \6.88 .
PayPal will charge the least fee, as shown above.
Specific behaviours
✓ calculates PayPal fee, using cap ✓ calculates Wise fee ✓ chooses company with least fee

- (b) Determine the amount, in Singapore dollars, his sister would receive if Jian used the most favourable exchange rate offered. (2 marks)

Solution
Wise has most favourable (highest) rate:
$1100 \times 0.8750 = 962.50$ Singapore dollars.
Specific behaviours
✓ indicates use of the highest rate ✓ correct amount of SGD

- (c) Jian's sister has told him that 1100 Australian dollars is not enough because she needs 1400 Singapore dollars. Determine the total funds, in Australian dollars, that Jian must have to afford this transfer using Wise, including the transfer fee. (3 marks)

Solution
AUD: $1400 \div 0.8750 = 1600$.
Transfer fee: $0.83 + 1600 \times 0.0055 = 9.63$
Total Jian will need is \$1609.63.
Specific behaviours
✓ uses exchange rate to obtain AUD amount ✓ shows correct expression for transfer fee ✓ correct total amount needed

Question 9

(9 marks)

- (a) A supermarket customer assistant was employed at a base rate of \$22.40 per hour. Calculate his total earnings in a week when he worked for 36 hours at his base rate and for 5 hours 30 minutes of over-time, for which he was paid time-and-a-half. (3 marks)

Solution
Overtime hours: $5.5 \times 1.5 = 8.25$
Total hours: $36 + 8.25 = 44.25$
Total earnings: $44.25 \times 22.40 = \$991.20$.
Specific behaviours
✓ adjusts overtime hours ✓ correct expression for base or overtime earnings using pay rate ✓ correct total earnings

- (b) Another person who ran an instore promotion for air fryers in the supermarket was paid a weekly retainer of \$245 plus 14.5% commission on the total value of the air fryers that the supermarket sold. Calculate the earnings for this person in a week when 26 small and 16 large air fryers were sold for \$119 and \$229 respectively. (3 marks)

Solution
Value of sales: $26 \times 119 + 16 \times 229 = 3094 + 3664 = 6758$.
Value of commission: $0.145 \times 6758 = 979.91$
Weekly earnings: $245 + 979.91 = \$1224.91$.
Specific behaviours
✓ indicates total value of sales ✓ correct expression to calculate commission ✓ correct weekly earnings

- (c) The supermarket sells the same yoghurt in two different sized cartons: 125 g for \$2.40 and 420 g for \$8.19. Use the unit cost method to compare the prices of each size carton and hence state which size represents the best value. (3 marks)

Solution
Unit costs: Small $240 \div 125 = 1.92$ cents per gram.
Large $819 \div 420 = 1.95$ cents per gram.
Small size is best value as the unit cost is less than for the large size.
Specific behaviours
✓ one unit cost ✓ both unit costs, with units shown ✓ correctly chooses best value size
Maximum 2 out 3 marks if did not use the unit cost method.

Question 10**(10 marks)**

Giovanna made some gelato and left it to cool in a cylindrical glass container with an internal diameter of 17 cm.

- (a) The container is filled with gelato to a uniform depth of 14 cm. Determine the volume of gelato that Giovanna has made, to the nearest cm^3 . (3 mark)

Solution
$V = \pi \times 8.5^2 \times 14$ $= 3178 \text{ cm}^3$ Volume of gelato is 3178 cm^3.
Specific behaviours
✓ Uses radius not diameter ✓ correctly substitutes into formula ✓ correct volume to nearest cm^3

Previously, Giovanna had made 2650 cubic centimetres of gelato and used it to fill right circular waffle cones with internal radius 2.5 cm and height 11 cm as treats for his friends.

- (b) Determine the volume of gelato required to fill one of these cones level with its circular top. (2 marks)

Solution
$V = \frac{1}{3}\pi(2.5)^2 \times 11$ $= 71.9948 \text{ cm}^3$ $= 72.0 \text{ cm}^3$ Volume of gelato to fill one cone is 72 cm^3.
Specific behaviours
✓ correctly substitutes into volume formula ✓ correct volume – accept either answer

- (c) How many complete cones could Giovanna fill? (1 mark)

Solution
Number of cones is $2650 \div 72 = 36.806$. Giovanna could fill 36 cones completely.
Specific behaviours
✓ correct number of cones

(d) What volume of gelato was left over?

(1 mark)

Solution
$V = 2650 - 36 \times 72$ $= 2650 - 2592$ $= 58 \text{ cm}^3$ 58 cm³ of gelato was left over.
Specific behaviours
✓ correct volume

Giovanna used the leftover gelato to make two types of small frozen shapes - cubes with a side length of 14 mm and spheres with a radius of 9 mm.

(e) Which shape used the least amount of gelato, and how much less gelato than the other shape?

(3 marks)

Solution	Alternate Solution
$V_C = 14^3$ $= 2744 \text{ mm}^3$ $V_S = \frac{4}{3}\pi(9)^3$ $= 3054 \text{ mm}^3$ $\Delta = 3054 - 2744$ $= 310 \text{ mm}^3$ The cube used the least amount of gelato, using 310 mm³ less than the sphere.	$V_C = 1.4^3$ $= 2.74 \text{ cm}^3$ $V_S = \frac{4}{3}\pi(0.9)^3$ $= 3.05 \text{ cm}^3$ $\Delta = 3.05 - 2.74$ $= 0.31 \text{ cm}^3$ The cube used the least amount of gelato, using 0.31 cm³ less than the sphere.
Specific behaviours	Specific behaviours
✓ volume of cube ✓ volume of sphere ✓ correctly states shape using least, and how much less, with units	✓ volume of cube ✓ volume of sphere ✓ correctly states shape using least, and how much less, with units

Question 11**(12 marks)**

Tamsin has prepared a weekly budget to check that she can afford to run a second-hand petrol fuelled car using the income from her casual job and government allowance.

Budget item	Weekly amount (\$)
Loan repayment	64
Registration and compulsory insurance	29
Comprehensive insurance	x
Petrol	45
Servicing	11
Roadside emergency assistance	3

(a) Name a budget item from the table above that is an example of

(i) a fixed expense.

(1 mark)

Solution
Loan repayment or Registration and compulsory insurance.
Specific behaviours
✓ names a correct fixed expense

(ii) a discretionary expense.

(1 mark)

Solution
Comprehensive insurance or Petrol or Servicing or Roadside emergency assistance.
Specific behaviours
✓ names a correct discretionary expense

(b) Assuming there are exactly 52 weeks in a year, determine

(i) the annual amount that Tamsin has budgeted for petrol.

(1 mark)

Solution
$A = 45 \times 52 = \$2340$
Specific behaviours
✓ correct amount

(ii) the value of x in the table (the weekly amount Tamsin has budgeted for comprehensive insurance) if her insurance policy costs \$78 per month.

(2 marks)

Solution
Annual amount: $78 \times 12 = \$936$.
Weekly amount: $x = 936 \div 52 = \$18$.
Specific behaviours
✓ indicates correct annual amount
✓ correct weekly amount

Tamsin is eligible for a maximum fortnightly Youth Allowance payment of \$425, but the actual amount she receives is reduced because she has a casual job from which she earns \$360 per week.

Youth Allowance payments reduce according to the following rules:

Amount your payment reduces by if you earn between \$480 and \$575 per fortnight	Amount your payment reduces by if you earn more than \$575 per fortnight
50 cents for each dollar over \$480	\$47.50 plus 60 cents for each dollar over \$575

- (c) Determine the Youth Allowance payment made to Tamsin each fortnight. (3 marks)

Solution
Fortnightly earnings: $2 \times 360 = \$720$.
Payment reduction: $47.50 + 0.60 \times (720 - 575) = 47.50 + 0.60 \times 145$ $= 47.50 + 87$ $= \$134.50$
Fortnightly payment will be $425 - 134.50 = \$290.50$
Specific behaviours
✓ indicates fortnightly earnings ✓ calculates payment reduction ✓ correct fortnightly payment

Tamsin cannot afford to spend more than 35% of her total earnings and allowance to run a car.

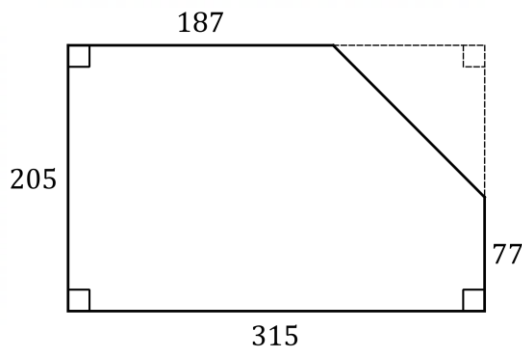
- (d) Will she have to find a way to reduce her budget? Justify your answer. (4 marks)

Solution
Total weekly income: $360 + 290.50 \div 2 = 360 + 145.25$ $= \$505.25$
Tasmin cannot spend more than $35\% \times 505.25 = \$176.84$.
Total of budget items: $64 + 29 + 18 + 45 + 11 + 3 = \170 .
Tasmin will not have to reduce her budget as her car expenses amount to \$170, which is less than 35% of her income.
Specific behaviours
✓ calculates total weekly income ✓ calculates maximum she can spend on car ✓ sums budget expenses ✓ correctly responds to question
Allow F/T from (c)

Question 12

(10 marks)

A manufacturer sells a range of rectangular timber-framed windows with a clipped corner. Four of the edge lengths of a piece of glass used to glaze one of these windows are shown in the diagram below, where the dashed line shows the triangular corner that is removed. The lengths are in centimetres and the diagram is not drawn to scale.



- (a) Determine the area of glass removed when the triangular corner is clipped. (3 marks)

Solution	
Base of triangle is $315 - 187 = 128$.	
Height of triangle is $205 - 77 = 128$.	
Area of triangle:	
	$A = \frac{1}{2} \times 128 \times 128$ $= 8192 \text{ cm}^2$
Specific behaviours	
✓ correctly calculates one dimension of triangle – can be given in the formula working ✓ shows correct use of area formula ✓ correct area – accept 0.82 m^2	

- (b) Determine the area of glass that remains once the corner has been clipped. (2 marks)

Solution
Area of rectangle:
$A_R = 205 \times 315$ $= 64\,575$
Area of glass remaining:
$A = 64\,575 - 8192$ $= 56\,383 \text{ cm}^2$
Specific behaviours
✓ calculates area of rectangle – accept 6.46 m^2 ✓ calculates remaining area – accept 5.64 m^2
Allow F/T from (a) if working shown.

- (c) Show use of Pythagoras' Theorem to determine the length of cut required to clip the corner. (2 marks)

Solution
Let x be length of cut, so that: $x^2 = 128^2 + 128^2$ $x^2 = 32\,768$ $x = \sqrt{32\,768}$ $= 181 \text{ cm}$
Specific behaviours
✓ clearly shows correct use of theorem in any form ✓ correct cut length – accept 1.81 m

Just before it is assembled in the timber frame, a thin layer of sealing tape is applied to the perimeter of the clipped pane of glass. The sealing tape costs 35 cents per metre.

- (d) Determine the cost of the sealing tape applied to the clipped pane of glass. (3 marks)

Solution
Perimeter of pane: $P = 187 + 205 + 315 + 77 + 181$ $= 965 \text{ cm}$ Cost of tape: $C = 965 \div 100 \times 0.35$ $= \$3.38$
Specific behaviours
✓ shows correct calculation for perimeter of pane ✓ shows correct calculation for cost of tape ✓ correct cost of tape to nearest cent – accept 338 cents

Question 13

(11 marks)

A financial advisor rents a small office to work in and meet clients.

- (a) A year ago, the annual rent payable for the office was set at \$460 per square metre. This amount is subject to an annual increase of 2.5% plus the percentage increase in the consumer price index over the previous year.

The consumer price index rose from 124.2 to 132.4 over the past year.

- (i) Show that the percentage increase in the consumer price index was 6.60%.

(1 mark)

Solution
% Increase in index = $\frac{132.4-124.2}{124.2} \times 100 = 6.60\%$.
Specific behaviours
✓ correctly substitutes into formula

Solution
Increase in index is $132.4 \div 124.2 = 1.06602$, which represents an increase of 6.60%.
Specific behaviours
✓ divides current index by previous

- (ii) Determine the new annual rent payable for the office given that it has a floor area of 62.5 square metres.

(3 marks)

Solution
Percentage rent increase will be $2.5 + 6.6 = 9.1\%$.
New rent per square metre is $460 \times 1.091 = \$501.86$.
New annual rent is $62.5 \times 501.86 = \$31\,366.25$.
Specific behaviours
✓ indicates correct total percentage increase
✓ calculates new rent per square metre
✓ correct new annual rent

- (b) A client of the financial advisor wanted a 60 day loan of \$1095. If the loan attracted simple interest at 12.5% per annum, determine the total amount that the client would have to repay at the end of the loan period.

(3 marks)

Solution
Simple interest: $I = \frac{1095 \times 12.5 \times 60}{100 \times 365} = 22.50$
Principal and interest: $1095 + 22.50 = 1117.50$
A sum of \$1117.50 must be repaid.
Specific behaviours
✓ correct expression for simple interest
✓ correctly evaluates simple interest
✓ correct sum to repay

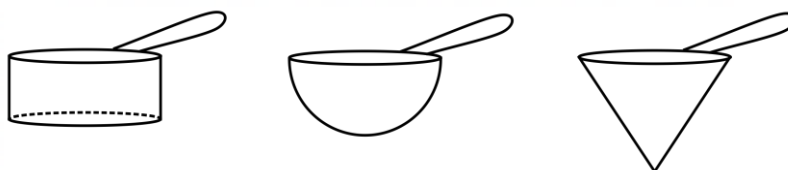
- (c) Another client was advised to invest a sum of \$22 500 for 18 months in a product that paid interest at a rate of 3.9% per annum compounded quarterly. Determine the total interest this investment would earn. (4 marks)

Solution
Future value of investment: $FV = 22\,500 \left(1 + \frac{3.9}{100 \times 4} \right)^{\frac{18}{12} \times 4}$ $= 22\,500 \times (1.00975)^6$ $= 23\,848.75$ Interest earned: $23\,848.75 - 22\,500 = \\1348.75
Specific behaviours
✓ correct expression for future value of investment ✓ correctly converts months to years ✓ correctly evaluates future value ✓ correct interest earned

Question 14

(10 marks)

A student has designed three metal scoops with identical wooden handles that can all be used by a baker to scoop exactly 150 grams of flour out of a large bin of the ingredient. They are shown in the diagram below, not drawn to scale.



One scoop is cylindrical with a radius of 4.5 cm and a height of 3 cm, one is hemispherical with a radius of 4.5 cm, and the other is conical with a radius of 4.5 cm and a perpendicular height of 9 cm.

- (a) Determine the area of metal required to make the cylindrical scoop. (2 marks)

Solution
Area of curved side: $A = \pi(4.5)^2 + 2\pi(4.5)(3)$ $= 63.62 + 84.82$ $= 148.42 \text{ cm}^2$
Specific behaviours
✓ indicates correct method - no lid ✓ correct area of metal

- (b) Determine the area of metal used to make the hemispherical scoop. (2 marks)

Solution
Area of metal for hemisphere: $A = 4\pi(4.5)^2 \times \frac{1}{2}$ $= 254 \times \frac{1}{2}$ $= 127.23 \text{ cm}^2$
Specific behaviours
✓ indicates correct method – no lid ✓ correct area of metal

Alternate Solution
Area of metal for hemisphere: $A = 2\pi(4.5)^2$ $= 127.23 \text{ cm}^2$
Specific behaviours
✓ indicates correct method – no lid ✓ correct area of metal

- (c) Determine the slant height of the conical scoop and hence state the area of metal required to make the conical scoop. (3 marks)

Solution
Slant height: $h = \sqrt{4.5^2 + 9^2}$ $= 10.06 \text{ cm}$
Area of metal for conical scoop: $A = \pi(4.5)(10.06)$ $= 142.22 \text{ cm}^2$
Specific behaviours
✓ correct slant height – accept 10.1 cm and 10 cm ✓ indicates correct method to calculate surface area – no lid ✓ correct area of metal If used 10.1 cm, accept $A = 142.29 \text{ cm}^2$ If used 10 cm, accept $A = 141.37 \text{ cm}^2$

See next page

- (d) By first calculating the volume of any one of the scoops, determine the volume of one gram of flour. (3 marks)

Solution
$V_{CYL} = \pi(4.5)^2(3) = 190.85 \text{ cm}^3$ $V_{HS} = \frac{4\pi(4.5)^3}{3} \times \frac{1}{2} = 190.85 \text{ cm}^3$ $V_{CONE} = \frac{\pi(4.5)^2(9)}{3} = 190.85 \text{ cm}^3$ Hence volume of one gram of flour is $190.85 \div 150 = 1.27 \text{ cm}^3$.
Specific behaviours
✓ shows correct use of formula for chosen scoop ✓ obtains correct volume ✓ correct volume of one gram

Question 15

(10 marks)

Five remote climate monitoring stations are numbered from 1 to 5. Due to their locations, not all stations can transmit data directly to the other stations. A list of all possible direct transmissions is given below.

- Station 1 can transmit data to Station 3
- Station 2 can transmit data to Station 3 and to Station 4
- Station 3 can transmit data to Station 2 and to Station 5
- Station 4 can transmit data to Station 1 and to Station 3
- Station 5 can transmit data to Station 4

Furthermore, no station can transmit data to itself, nor can it ever receive its own data back.

- (a) Construct matrix N to represent this transmission network, with stations in numerical order and with element $n_{i,j}$ equal to 1 if station i can transmit data to station j , and equal to 0 if not. (2 marks)

Solution	
$N =$	$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$
Specific behaviours	
✓ matrix containing eight 1's and leading diagonal with only 0's	
✓ correct matrix	

- (b) Determine matrix M , where $M = N^2$. (2 marks)

Solution	
$M = N^2 =$	$\begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \end{bmatrix}$
Specific behaviours	
✓ at least 2 correct rows or columns	
✓ correct matrix	

- (c) State the value of element $m_{2,2}$ in matrix M and explain why it contradicts one of the conditions in the introduction to this question. (2 marks)

Solution	
$m_{2,2} = 1$ which means that Station 2 can transmit data to itself via one other station, but no station can ever receive its own data back.	
Specific behaviours	
✓ states value	
✓ explains contradiction	

- (d) Replace all the ones on the leading diagonal of matrix M with zeros, to form matrix P . (1 mark)

Solution					
$P = \begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \end{bmatrix}$					
Specific behaviours					
✓ correct matrix					

- (e) Determine matrix $Q = N + P$. (1 mark)

Solution					
$Q = N + P = \begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 2 & 1 & 1 \\ 0 & 1 & 0 & 2 & 1 \\ 1 & 1 & 2 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \end{bmatrix}$					
Specific behaviours					
✓ correct matrix					

- (f) Use matrix Q to state, with justification, which station(s) can transmit data to all other stations either directly or via one other station. (2 marks)

Solution	
Stations 2 and 4.	
Rows 2 and 4 of matrix Q are the only rows in which all elements (apart from those on the leading diagonal) are not zero.	
Specific behaviours	
✓ correctly identifies both stations	
✓ supplies justification that reflects correct interpretation of matrix elements	

Question 16

(10 marks)

Danielle's bedroom is a rectangular room that measures 4.65 m by 6.35 m and she has created a scale drawing of it using a scale of 1: 25.

- (a) Would her drawing fit on a 20 cm by 25 cm sheet of paper? Justify your answer.

(3 marks)

Solution
$465 \div 25 = 18.6 \text{ cm}, \quad 635 \div 25 = 25.4 \text{ cm}$
Drawing would not fit – the paper is wide enough but not quite long enough.
Specific behaviours
✓ correct division of room size by 25 seen ✓ two correct dimensions for scaled room in cm ✓ states drawing would not fit, with reason

Solution
$20 \times 25 \div 100 = 5 \text{ m}, \quad 25 \times 25 \div 100 = 6.25 \text{ m}$
Drawing would not fit – the paper is wide enough but not quite long enough.
Specific behaviours
✓ correct multiplication of paper size by 25 seen ✓ two correct dimensions for scaled paper in m ✓ states drawing would not fit, with reason

- (b) On the floor of her bedroom she has a circular rug with a radius of 1.05 m. It is split into different coloured sectors, the green one having a central angle of 55° . Determine

- (i) the area of the green sector of the actual rug.

(2 marks)

Solution
Area of sector is $\pi \times 1.05^2 \times \frac{55}{360} = 0.529 \text{ m}^2$.
Specific behaviours
✓ correct fraction of whole circle ✓ correct area of sector

- (ii) the area, in square centimetres, of the green sector on her scale drawing of the rug.

(2 marks)

Solution
Area scale factor is $1: 25^2 = 1: 625$.
Area of rug on drawing is $0.529 \div 625 \times 100^2 = 8.46 \text{ cm}^2$.
Specific behaviours
✓ indicates correct area scale factor ✓ correct area of rug

- (c) A cube of foam in her bedroom that her friends sit on needs a new cover. On her scale drawing, the length of one side of the cube is 22 mm. Allowing an extra 20% for seams and wastage, how many square centimetres of fabric should be bought to make a new cover for the cube?

(3 marks)

Solution
Actual side length of cube is $2.2 \times 25 = 55 \text{ cm}$.
Total surface area of cube is $6 \times 55^2 = 18\,150 \text{ cm}^2$.
Fabric required is 20% more: $18\,150 \times 1.20 = 21\,780 \text{ cm}^2$.
Specific behaviours
✓ correct side length of cube ✓ correct TSA for cube ✓ correct area of fabric, with units in square cm

See next page

Question 17

(8 marks)

A borrower must make n monthly repayments of A dollars to repay a loan of L dollars when the interest rate on their loan is $R\%$ per annum compounded monthly. To calculate the monthly repayment A , the following formulas are used:

$$A = \frac{Lr}{b}, \text{ where } r = \frac{R}{1200} \text{ and } b = 1 - (1 + r)^{-n}.$$

- (a) Lars has negotiated a loan of \$52 000 to buy a new EV. The interest rate on his loan is 7.8% per annum compounded monthly and he will make 60 monthly repayments.

- (i) Calculate the value of r for this loan.

(1 mark)

Solution
$r = 7.8 \div 1200 = 0.0065$
Specific behaviours
✓ correct value of r

- (ii) Calculate the value of b , rounding your answer to 6 decimal places.

(2 marks)

Solution
$b = 1 - (1 + 0.0065)^{-60} = 0.322\ 088$
Specific behaviours
✓ correct ' n ' value (-60)
✓ correctly calculates value of b to 6 dp

- (iii) Show that the monthly repayment Lars must make is \$1049.40.

(1 mark)

Solution
$A = 52\ 000 \times 0.0065 \div 0.322\ 088 = 1049.40.$
Lars must repay \$1049.40 each month.
Specific behaviours
✓ shows correct calculation for A

- (b) Leigh has budgeted monthly repayments of no more than \$150 to buy an electric scooter priced at \$3250. She has been offered a loan for the full purchase price with 24 monthly repayments at an interest rate of 8.4% per annum compounded monthly. If Leigh goes ahead with the loan, will the repayments satisfy her budget? Justify your answer.

(4 marks)

Solution
$r = 8.4 \div 1200 = 0.007$
$b = 1 - (1 + 0.007)^{-24} = 0.154151$
$A = 3250 \times 0.007 \div 0.154151 = 147.58$
The monthly repayment will be \$147.58 and as this is less than her budgeted amount of \$150 the repayments will satisfy her budget.
Specific behaviours
✓ indicates correct value of r
✓ indicates correct value of b
✓ states correct monthly repayment to the nearest cent
✓ correctly responds to question about budget

Supplementary page

Question number: _____

Supplementary page

Question number: _____

Supplementary page

Question number: _____

Supplementary page

Question number: _____

